THE REDUCTION OF A SMOOTH REPRESENTABLE MORPHISM OF FORMAL ALGEBRAIC SPACES IS THE BASE CHANGE OF THE REDUCTION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The first sentence of the proof is a parenthetical referring to the construction of the reduction in an earlier proof; it is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The reduction of a formal algebraic space and its universal property. Same chapter, Lemma lemma-reduction-formal-algebraic-space, invoked at proof lines 2930-2931 and 2937-2939. Supplies the functor whose universal property identifies the two sides.
- The base change of a smooth morphism is smooth and its source is reduced when the target is. Descent on Spaces, Lemma spaces-descent-lemma-reduced-local-smooth, invoked at proof lines 2932-2936. Makes the base changed reduction a reduced algebraic space.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GB3: sources/stacks-chow/tags.txt, source lines 20358–20358. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over S which is representable by algebraic spaces and smooth (for example étale). Then $X_{red} = X \times_Y Y_{red}$.*

Proof. (The étale case follows directly from the construction of the underlying reduced algebraic space in the proof of Lemma 0AIN.) Assume f is smooth. Observe that $X \times_Y Y_{red} \to Y_{red}$ is a smooth morphism of algebraic spaces. Hence $X \times_Y Y_{red}$ is a reduced algebraic space by Descent on Spaces, Lemma 0GB3. Then the universal

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property of reduction shows that the canonical morphism $X_{red} \to X \times_Y Y_{red}$ is an isomorphism. $\square$